

Performance Metrics

Used in Linear Regression

R-squared & Adjusted R-squared — Complete Notes with Detailed Explanation

(Har formula, har notation, aur har chhoti term ko asaan bhasha mein samjhaya gaya hai)

1. Introduction — Yeh Metrics Kyun Chahiye?

Jab hum Linear Regression ka model banate hain (ek Best Fit Line jo data ko predict karti hai), toh sirf line bana dena kaafi nahi hai. Humein yeh bhi pata hona chahiye ki hamara model kitna 'accurate' hai — matlab woh actual real-world data ko kitne achhe se represent kar raha hai.

Isi cheez ko measure karne ke liye kuch Performance Metrics use hote hain. Notes mein do important metrics cover kiye gaye hain:

- R-squared (R^2) — Model ki accuracy ka basic measure.
- Adjusted R-squared — R^2 ki ek improved version, jo ek important problem ko fix karta hai.

In dono ko samajhne se pehle, hum unn building-block terms ko samajhte hain jinse yeh formulas bante hain.

2. Basic Building Blocks — y_i , $\hat{y}_i$, $\bar{y}$

R^2 ka formula teen chhoti-chhoti values par based hai. In teeno ko clearly samajhna bahut zaroori hai, kyunki yehi poore formula ki neev (foundation) hain:

- y_i (y actual) — Dataset mein diya gaya i-th data point ka Actual / True value. Yeh woh real value hai jo humein already pata hai (jaise actual height, actual price, etc.).
- $\hat{y}_i$ (y-hat, predicted) — Hamare model (Best Fit Line) ne i-th data point ke liye jo value Predict ki hai. Line par us x-value ke corresponding jo point hai, wahi $\hat{y}_i$ hai.
- $\bar{y}$ (y-bar, average of y) — Saare actual y values ka Mean/Average. Yeh ek single constant value hai (poore dataset ke liye same rehta hai), aur graph mein yeh ek horizontal dotted line ke roop mein dikhaya jaata hai.

Graph mein teeno cheezein is tarah dikhti hain: data points (x marks) ke around ek diagonal Best Fit Line hoti hai (jispar $\hat{y}_i$ values hoti hain), aur ek horizontal dotted line hoti hai jo $\bar{y}$ (average) ko represent karti hai. Actual point aur line ke beech ki vertical distance ko Residual ya Error kehte hain.

Performance Metrics Used In Linear Regression

① R squared

$$R_{\text{squared}} = 1 - \frac{SS_{\text{Res}}}{SS_{\text{Total}}}$$

$$= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2}$$

$$= 1 - \frac{\text{Small number}}{\text{Bigger number}}$$

$$= 1 - \text{Smaller number}$$

$$\approx 1$$

$$0.70 \Rightarrow 70\%$$

$$0.85 \Rightarrow 85\%$$

$$0.90 \Rightarrow 90\%$$

{ Overfitting, Underfitting }

→ 1

↓

Accuracy

Model Is?

R squared ↑↑↑

Size of the house ↑ Price ↑

the correlation

No. of bedrooms ↑ Price ↑

the correlation

② Adjusted R squared

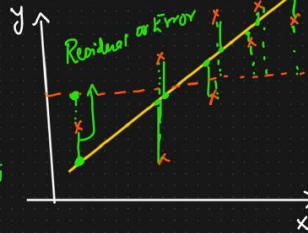
Dataset

→ (Price)

Gender | Size of the house | No. of bedrooms | Location | Price

$$R_{\text{squared}} = 75\% \Rightarrow 0.75$$

This is the problem of R squared



Isme dikhaya gaya hai: R squared aur Adjusted R squared ka intro, R^2 ka formula ($SS_{\text{res}}/SS_{\text{total}}$), graph mein Residual/Error, $\hat{y}_i$ aur $\bar{y}_i$ (average line), aur Adjusted R^2 ki zaroorat kyun padi (Gender jaise irrelevant feature ka example).

3. SS_{res} aur SS_{total} — R^2 ke Do Main Ingredients

SS_{res} — Sum of Squares of Residuals

SSres batata hai ki hamara Model kitna 'galat' hai — matlab predicted values ($\hat{y}_i$), actual values (y_i) se kitni door hain.

$$SSres = \sum (y_i - \hat{y}_i)^2$$

Yahan har data point ke liye Actual value minus Predicted value nikala jaata hai (yeh Residual/Error hai), fir use square kiya jaata hai (taaki negative-positive cancel na ho), aur fir saare data points ke liye add ($\sum$) kar diya jaata hai. Simple shabdon mein — yeh humara Model ka total error hai.

SSTotal — Total Sum of Squares

SSTotal batata hai ki actual values apni khud ki Average ($\bar{y}_i$) se kitni door hain — matlab agar hum koi bhi model use hi na karte aur sirf har baar average value predict kar dete, toh kitna error hota.

$$SSTotal = \sum (y_i - \bar{y}_i)^2$$

Yeh ek tarah ka 'Baseline' error hai — ek bahut hi simple (dumb) model ka error, jo bina kisi pattern seekhe sirf average predict karta hai.

Note: SSres aur SSTotal dono ka formula dikhne mein bilkul similar hai — bas dusri term alag hai. SSres mein hum Predicted ($\hat{y}_i$) se compare karte hain, jabki SSTotal mein hum Average ($\bar{y}_i$) se compare karte hain.

4. R-squared (R^2) ka Formula

Ab in dono values ko combine karke R^2 ka formula banta hai:

$$R^2 = 1 - (SSres / SSTotal)$$

Poora expanded formula is tarah likha jaata hai:

$$R^2 = 1 - [\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y}_i)^2]$$

Yeh Formula Kaam Kaise Karta Hai — Intuition

Is formula ko step-by-step samjhte hain:

- $SSres / SSTotal$ ek ratio (fraction) hai — 'Hamare Model ka Error' ko 'Baseline (Average) Model ka Error' se divide kiya gaya hai.
- Agar hamara Model achha hai, toh uska error (SSres) bahut kam hoga baseline error (SSTotal) ke comparison mein. Isliye yeh fraction ek Small Number banega.
- 1 minus ek chhoti si value = ek value jo 1 ke bahut kareeb hoti hai. Isliye achhe model ke liye $R^2 \approx 1$ hota hai.

$$R^2 = 1 - (\text{Small Number} / \text{Bigger Number}) = 1 - \text{Smaller Number} \approx 1$$

Agar model bura hai, toh SSres, SSTotal ke kareeb-kareeb barabar ho jaayega (matlab model, average predict karne se zyada behtar nahi hai), aur fraction ki value 1 ke kareeb ho jaayegi, jisse R^2 ki value 0 ke kareeb ho jaayegi.

R^2 Ki Values Ko Kaise Interpret Karein

R^2 generally 0 se 1 ke beech ek value hoti hai, jise percentage mein bhi likha ja sakta hai:

R^2 (decimal)	R^2 (percentage)	Matlab
0.70	70%	Model, data ki 70% variation ko explain kar pa raha hai

0.85	85%	Model, data ki 85% variation ko explain kar pa raha hai
0.90	90%	Model, data ki 90% variation ko explain kar pa raha hai

Simple shabdon mein, R^2 yeh sawaal answer karta hai: 'Accurately my model is?' — matlab hamara model, actual data ke pattern ko kitne percent tak sahi-sahi capture kar paa raha hai.

Note: R^2 ka high hona hamesha 'perfect model' guarantee nahi karta — kabhi kabhi bahut high R^2 Overfitting ka sign bhi ho sakta hai, aur bahut low R^2 Underfitting ka sign hota hai. Dono hi cases mein model real-world data par sahi se perform nahi karta.

5. R-squared Ki Ek Badi Problem

R^2 ek simple aur useful metric hai, lekin isme ek important khaami (flaw) hai — jaise-jaise hum apne dataset mein zyada Independent Features (input columns) add karte jaate hain, R^2 ki value hamesha badhti hi jaati hai (ya same rehti hai), chahe woh naya feature actually useful ho ya na ho.

Isse ek example se samajhte hain. Maan lijiye humein House Price predict karna hai, aur hamare paas yeh features hain:

Dataset →	Predict karna hai
Size of the house, No. of bedrooms, Location	Price

Is dataset ke saath model banane par maan lijiye $R^2 = 75\%$ (0.75) aata hai.

Logical (Positive) Correlations

- Size of the house $\uparrow \rightarrow$ Price $\uparrow$ (True correlation — bade ghar ki price zyada hoti hai, yeh sensible hai).
- No. of bedrooms $\uparrow \rightarrow$ Price $\uparrow$ (True correlation — zyada bedrooms wale ghar ki price zyada hoti hai, yeh bhi sensible hai).

In dono features ka Price ke saath genuine relationship hai, isliye inka model mein hona sahi hai.

Ab Ek Irrelevant Feature Add Karte Hain

Ab agar hum dataset mein ek aur column add kar dein — 'Gender' (jo ki house price predict karne ke liye logically kisi kaam ka nahi hai) — toh bhi R^2 ki value badh jaayegi (jaise 75% se 80%, phir 85%, phir 87% — jaisa notes mein dikhaya gaya hai).

Zyada Features Add Karo $\Rightarrow R^2$ Hamesha $\uparrow\uparrow\uparrow$ (Kabhi Kam Nahi Hota)

Yeh problematic hai, kyunki isse humein galat impression milta hai ki model 'better' ho raha hai, jabki hakikat mein hum sirf random/irrelevant features add kar rahe hain jo model ko sirf training data ke saath zyada 'fit' kar rahe hain (bina koi real predictive value diye).

Yehi hai R-squared ki main problem: R^2 kabhi bhi feature add karne par decrease nahi hota, chahe woh feature useless hi kyun na ho. Isi problem ko fix karne ke liye Adjusted R-squared banaya gaya.

6. Adjusted R-squared

Adjusted R^2 bhi R^2 jaisa hi ek metric hai, lekin yeh ek extra factor add karta hai jo Number of Features ko bhi dhyan mein rakhta hai. Agar koi naya feature model ko genuinely improve nahi kar raha, toh Adjusted R^2 ki value R^2 ke mukable Kam ho jaayegi (ya badhne ke bajaye ghat bhi sakti hai) — isse humein pata chal jaata hai ki naya feature actually useful tha ya nahi.

$$\text{Adjusted } R^2 = 1 - [(1 - R^2)(N - 1) / (N - P - 1)]$$

Formula Ke Har Symbol Ka Matlab

- R^2 — Wahi normal R-squared value jo humne upar calculate ki (0 se 1 ke beech).
- N — Dataset mein total Data Points (rows) ki sankhya, matlab total training examples.
- P — Dataset mein total Independent Features (predictors/input columns) ki sankhya — jaise Size, Bedrooms, Location, etc.
- $(N - 1)$ — Yeh 'degrees of freedom' ka ek adjustment hai total data ke liye.
- $(N - P - 1)$ — Yeh model ke residual (error) degrees of freedom ko represent karta hai — jitne zyada features (P) honge, yeh value utni hi chhoti hoti jaayegi.

Formula ka core idea yeh hai: jaise-jaise P (features ki sankhya) badhta hai, denominator $(N - P - 1)$ chhota hota jaata hai, jisse poora fraction $[(1-R^2)(N-1)/(N-P-1)]$ bada ho jaata hai — aur isse Adjusted R^2 ki value neeche aa jaati hai, jab tak ki naya feature R^2 ko itna zyada na badha de ki yeh penalty cover ho jaaye.

Matlab — Adjusted R^2 , naye feature ko ek 'penalty' deta hai. Agar feature genuinely useful hai (aur R^2 ko kaafi badha deta hai), tab hi Adjusted R^2 badhega. Agar feature useless hai, toh penalty ki wajah se Adjusted R^2 gir jaayega.

$$R^2_{\text{squared}} \Rightarrow 80\% \Rightarrow 0.80$$

$$R^2_{\text{squared}} \Rightarrow 85\% \Rightarrow 0.85$$

$$R^2_{\text{squared}} \Rightarrow 87\% \Rightarrow 0.87$$

$$\text{Adjusted } R^2_{\text{squared}} = 1 - \frac{(1-R^2)(N-1)}{N-p-1} \quad \left\{ \begin{array}{ll} p=2 & R^2=90\% \quad R^2_{\text{adjusted}}=86\% \\ p=3 & R^2=92\% \quad R^2_{\text{adjusted}}=82\% \end{array} \right.$$

N = No. of data points

p = No. of Independent features

Isme dikhaya gaya hai: R^2 ki values (80%, 85%, 87%) jaise-jaise features add hote gaye, Adjusted R^2 ka poor formula, aur do examples ($P=2$ aur $P=3$) jinme R^2 badhne ke bawajood Adjusted R^2 kam ho jaata hai.

7. Worked Example — R^2 vs Adjusted R^2

Notes mein diye gaye do examples se yeh concept aur clear ho jaata hai:

Features (P)	R ²	Adjusted R ²
P = 2	90%	86%
P = 3	92%	82%

Dekha jaaye toh jab hum P=2 se P=3 kiya (ek naya feature add kiya), toh R² thoda sa badh gaya (90% se 92%) — jo yeh dikhata hai ki naya feature 'kuch' improvement laaya. Lekin Adjusted R² ulta ghat gaya (86% se 82%)!

Iska matlab hai — us naye (teesre) feature ne jitna R² badhaya, uske comparison mein utni value add nahi ki jitni ek 'genuinely useful' feature ko karni chahiye thi. Isliye Adjusted R² ne is feature ko penalize kar diya aur bata diya ki yeh feature zaroori nahi tha.

Isi wajah se: Real-world Machine Learning practice mein, jab hum multiple features waale (Multiple Linear Regression) models compare karte hain, tab sirf R² dekhna kaafi nahi hota — Adjusted R² dekhna zyada reliable hota hai, kyunki yeh unnecessary features ke against protect karta hai.

8. Summary — Dono Metrics Ka Farak

Point	R-squared	Adjusted R-squared
Kya measure karta hai	Model, data ki kitni variation explain karta hai	R ² ka improved version, features ki sankhya ke saath adjusted
Naya feature add karne par	Hamesha badhta hai (ya same rehta hai)	Sirf tabhi badhta hai jab feature genuinely useful ho
Formula	$1 - (SS_{res}/SS_{Total})$	$1 - [(1-R^2)(N-1)/(N-P-1)]$
Use kab karein	Single feature (Simple Linear Regression) models ke liye	Multiple features (Multiple Linear Regression) models compare karne ke liye

Dono metrics ka final goal same hai — yeh check karna ki hamara Linear Regression model kitna accurate hai — bas Adjusted R² zyada fair aur reliable measure deta hai jab dataset mein multiple features involved hon.